

AGRI COMMODITY PRICE FORECASTING

& MARKET INTELLIGENCE DASHBOARD

CASE STUDY REPORT

Project	Domain	Data Sources	Period
Agri Market Intelligence	Agricultural Data Science	Agmarknet · eNAM · Govt Open Data	2025–26 Season

Tech Stack	Models	Output	Status
Python · SQL · Power BI · Excel	SARIMA · Prophet · XGBoost	Dashboard + Forecast + Reports	Active

1. Executive Summary

Commodities Tracked	Avg Modal Price	Total Arrivals	Forecast Accuracy	MSP Crops Covered
95 <i>Karnataka APMCs</i>	₹4,821 <i>+2.3% MoM</i>	2.14L MT <i>22-Jun-2026</i>	91.4% <i>MAPE 8.6%</i>	24 <i>2025–26 Season</i>

India's agricultural marketing system processes over **7,000+ daily mandi transactions** across state APMCs, yet price discovery remains opaque for most farmers. This project builds an end-to-end **Agri Commodity Price Forecasting & Market Intelligence Dashboard** using real government data from Agmarknet, eNAM, and FAOSTAT to solve three key problems:

Business Problem	Solution Delivered
• Farmers lack real-time price signals across mandis • Price forecasting requires costly intermediaries • MSP procurement gaps go undetected until too late • Seasonal patterns are invisible to smallholders	• Live mandi price tracker across 42 Karnataka APMCs • 30-day ensemble price forecast (SARIMA + Prophet + XGB) • MSP risk alerts with procurement intervention signals • Monthly seasonal index for 16 major crops

2. Problem Statement

2.1 Market Context

Agriculture contributes **17% of India's GDP** and employs over 54% of the workforce. Yet agricultural marketing is plagued by information asymmetry. The **Agmarknet network** covers 3,000+ markets and generates daily price data, but this data remains difficult to access, analyse, and act upon — especially for price forecasting and MSP compliance monitoring.

2.2 Key Pain Points

#	Pain Point	Impact	Stakeholder
1	Price opacity across mandis — same crop ₹2,200–₹7,300 on same day	Farmer loses 20–35% potential income	Farmers / FPOs
2	No early-warning system for crops trading below MSP	Delayed procurement → distress sales	Policy makers
3	Seasonal pattern invisible — no lean-season premium capture	Missed storage arbitrage opportunity	Traders / Aggregators
4	30-day price forecast unavailable to most stakeholders	Poor sell-timing decisions	All participants
5	State-wise price disparity untracked	Inefficient inter-state trade routing	Wholesalers

3. Data Sources & Methodology

3.1 Datasets Used

Source	Dataset	Records	Frequency	Coverage
Agmarknet	Market-wise Daily Report	729 records	Daily	42 Karnataka APMCs, 95 commodities
Agmarknet	Season-wise Price & Arrival	48 rows	Seasonal	24 MSP crops, 2025–26 season
Agmarknet	State-wise Wholesale Monthly	12 states	Monthly	6 states, Arecanut, Jan 2025
CACP / Govt	MSP Rates 2025–26	24 crops	Annual	Kharif + Rabi commodities
FAOSTAT	Global benchmark prices	Reference	Annual	Wheat, Rice, Maize benchmark

3.2 Data Pipeline Architecture

Ingestion Layer - agmarknet_parser.py — handles irregular CSV structure - Stateful parser tracks state/market context per line 3 report types with different schemas handled - Data validation: price bounds, unit normalization	Processing Layer - cleaner.py — outlier removal, name standardisation - Commodity alias table maps 40+ Agmarknet variants - MSP join adds premium/discount flag per commodity - Lag features: 1d, 7d, 14d, 30d for ML models
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3.3 Forecasting Models

Model	Best For	MAPE	Key Config
SARIMA(1,1,1)(1,1,0)12	Cereals — stable seasonal patterns	6–7%	ADF test → d=1, s=12 monthly
Prophet	Pulses — trend shifts, policy floors	7–9%	changepoint_prior=0.05, yearly seasonal
XGBoost	Vegetables — high day-to-day volatility	11–13%	lag features + seasonal index, 5-fold CV
Ensemble (Weighted Avg)	All crops — balanced accuracy	~8.6%	Weights = inverse MAPE on 30d validation

4. Key Findings & Insights

01

Paddy: 4.7% above MSP — fair value confirmed

Paddy (Common) modal price ₹2,481 exceeds MSP ₹2,369 by 4.7% across Karnataka mandis. Farmers in Shimoga, Mandya, and Tumkur mandis are receiving above-floor prices. MSP procurement pressure remains low for 2025-26 Kharif season.

02

Groundnut: 12.2% below MSP — intervention trigger

Groundnut market price ₹6,376 is 12.2% below MSP ₹7,263. With 55,000 MT of arrivals, this represents ₹49.5 crore in potential income loss to farmers. NAFED procurement intervention recommended before Dec 2025.

03

Maharashtra Arecanut +39.3% YoY — supply shock signal

Maharashtra registered the highest wholesale price gain for Arecanut at ₹66,147/qtl (Jan 2025 vs Jan 2024: ₹47,500). The 39.3% YoY jump is driven by supply shortage from 2023-24 drought lag and rising export demand from South-East Asia.

04

Arrival-price correlation = -0.42 — inverse relationship confirmed

Across 95 commodities in Karnataka mandis, the Pearson correlation between daily arrivals and modal prices is -0.42. High-arrival crops (Coconut: 1.33L units) show price compression while low-arrival premium crops (Copra: ₹23,000/qtl) confirm supply-demand fundamentals.

05

Seasonal amplitude: Tomato ±56%, Arecanut ±15%

STL decomposition reveals Tomato has the widest seasonal price swing — lean-season peak (Aug) is 56% above harvest-season trough (Apr). Arecanut and Copra show the most stable seasonal patterns, making them suitable for year-round procurement planning.

06

Bajra: 16% below MSP — silent distress crop

Bajra (Pearl Millet) market price ₹2,331 is 16% below MSP ₹2,775. With 18,000 MT Kharif arrivals, this is the most underperforming MSP crop in the dataset. Concentrated in Rajasthan and UP — state-level procurement data needed for full risk assessment.

5. Market Intelligence Results

5.1 Current Mandi Prices — 22 Jun 2026 (Selected)

Commodity	Market	Min ₹	Max ₹	Modal ₹	Arrivals	MSP ₹	Status
Paddy	Shimoga APMC	2,350	2,650	₹2,481	320 MT	2,369	Above MSP
Wheat	Arasikere APMC	2,825	3,300	₹2,825	8 MT	2,425	Above MSP
Maize	Byadagi APMC	1,900	2,289	₹2,245	25 MT	2,400	Below MSP
Groundnut	Byadagi APMC	4,089	8,109	₹7,018	57 MT	7,263	Below MSP
Red Gram	Arasikere APMC	7,281	10,725	₹7,310	14 MT	8,000	Below MSP
Copra	Shimoga APMC	18,000	23,000	₹23,000	28 MT	12,100	Strong (+90%)
Cotton	Dharwad APMC	7,000	8,200	₹7,435	105 MT	7,710	Watch (-3.6%)
Mustard	Gadag APMC	5,800	6,800	₹6,114	88 MT	5,950	Above MSP

5.2 State-wise Arecanut Price Analysis

State	Jan 2025 ₹/qtl	Dec 2024 ₹/qtl	Jan 2024 ₹/qtl	MoM %	YoY %	Rank
Maharashtra	₹66,147	₹63,637	₹47,500	+3.9%	+39.3%	#1
Karnataka	₹37,820	₹37,979	₹40,046	-0.4%	-5.6%	#2
Keralam	₹33,645	₹33,514	₹36,061	+0.4%	-6.7%	#3
Goa	₹31,949	₹29,758	₹32,585	+7.4%	-1.9%	#4
Meghalaya	₹12,123	₹11,510	₹16,278	+5.3%	-25.5%	#5
Weighted Avg	₹30,714	₹29,780	₹26,342	+3.1%	+16.6%	—

6. Price Forecast — 30-Day Outlook

6.1 7-Day Forecast Summary (Jun 23 – Jun 29, 2026)

Commodity	Current ₹	Forecast ₹ (7d)	Change	Lower CI	Upper CI	Model MAPE	Signal
Paddy	₹2,481	₹2,476	-0.2%	₹2,435	₹2,517	6.2%	Bearish — kharif sowing
Wheat	₹2,488	₹2,492	+0.2%	₹2,454	₹2,530	5.8%	Neutral — stable MSP floor
Maize	₹2,600	₹2,591	-0.3%	₹2,533	₹2,649	8.4%	Bearish — ethanol demand
Tomato	₹1,500	₹1,515	+1.0%	₹1,353	₹1,677	12.4%	Bullish — lean season
Onion	₹2,200	₹2,196	-0.2%	₹2,088	₹2,304	11.1%	Neutral — kharif arrivals
Mustard	₹6,114	₹6,120	+0.1%	₹6,029	₹6,211	7.3%	Bullish — import parity
Cotton	₹7,435	₹7,433	-0.03%	₹7,336	₹7,530	8.9%	Neutral — below MSP ₹7,710
Copra	₹23,000	₹23,011	+0.05%	₹22,658	₹23,364	9.8%	Bullish — supply tight

6.2 Model Performance Summary

Crop	Model	MAPE	RMSE (₹)	Directional Acc.	Verdict
Paddy	SARIMA(1,1,1)(1,1,0)12	6.2%	₹148	72%	Excellent
Wheat	Prophet	5.8%	₹138	74%	Excellent
Maize	Ensemble	8.4%	₹215	68%	Good
Red Gram	Prophet	7.6%	₹540	66%	Good
Mustard	Prophet	7.3%	₹415	69%	Good
Cotton	ARIMA(2,1,2)	8.9%	₹652	65%	Acceptable
Tomato	XGBoost + Lags	12.4%	₹320	60%	Acceptable*

Onion	SARIMA + Exog	11.1%	₹280	62%	Acceptable*
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* High-volatility crops; accuracy improves with rainfall data as exogenous variable.

7. Seasonal Price Index

Seasonal indices (base = 100) were computed via **STL decomposition** on historical Agmarknet monthly data. An index above 100 indicates a lean-season price premium; below 100 indicates harvest-season surplus pricing.

Crop	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Peak
Paddy	78	75	82	90	105	118	115	108	95	85	82	78	Jun
Wheat	82	80	78	85	95	108	115	115	108	100	92	85	Jul
Tomato	95	90	85	80	82	92	115	125	118	105	98	92	Aug
Onion	90	88	85	82	85	95	112	122	115	102	95	90	Aug
Mustard	82	80	78	82	90	105	115	118	112	102	92	85	Aug
Cotton	85	82	80	78	82	95	108	115	112	100	92	88	Aug

Green (≥ 110) = lean season premium Red (≤ 88) = harvest surplus Amber column = peak month

8. SQL & Database Design

The project uses a **PostgreSQL 14+** database with 8 normalized tables, 3 materialized views, and 2 stored procedures. Below are the key query patterns used in the Power BI data model.

8.1 Core Query — MSP Risk Detection

```
-- MSP Risk Report: crops trading below minimum support price
SELECT c.name AS commodity, mr.season,
       mr.msp_value, ROUND(AVG(mp.modal_price),0) AS market_price,
       ROUND((AVG(mp.modal_price) - mr.msp_value) / mr.msp_value * 100, 1) AS
gap_pct,
       CASE WHEN AVG(mp.modal_price) < mr.msp_value * 0.95 THEN 'HIGH RISK'
            WHEN AVG(mp.modal_price) < mr.msp_value THEN 'WATCH' ELSE 'OK' END
AS risk
FROM   mandi_prices mp
JOIN   commodities c ON mp.commodity_id = c.id
JOIN   msp_rates mr  ON mr.commodity_id = c.id AND mr.marketing_year = '2025-26'
GROUP BY c.name, mr.season, mr.msp_value HAVING AVG(mp.modal_price) <
mr.msp_value;
```

8.2 Schema Overview

Table	Type	Rows (Est)	Key Columns	Purpose
commodity_groups	Reference	9	id, name, description	Group classification
commodities	Reference	33	id, name, group_id, is_msp_crop	Commodity master
markets	Reference	18+	id, name, state_id, district	APMC market registry
mandi_prices	Fact	729 / day	trade_date, market_id, commodity_id, modal_price, arrivals	Core price & arrival data
msp_rates	Reference	24	marketing_year, commodity_id, season, msp_value	MSP by crop/season
state_wholesale_prices	Fact	~100/month	price_month, state_id, commodity_id, avg_price, mom/yoy	State monthly analysis
season_arrivals	Aggregate	24	marketing_year, commodity_id, season, avg_price, msp_premium	Seasonal summary
forecasts	Fact	Growing	run_date, commodity_id, forecast_date, predicted_price, ci	Model output store

9. Power BI Dashboard Setup Guide

9.1 Data Model

The Power BI data model uses the Excel workbook **Agri_MarketIntel_PowerBI_Data.xlsx** (8 sheets) as the primary data source. Each sheet maps to one **Power BI table**.

Excel Sheet	Power BI Table	Role	Key Relationships
Mandi_Prices	fact_mandi_prices	Fact	→ Commodity, Market (by name)
Monthly_Price_Trend	fact_monthly_trend	Fact	→ Commodity name, Month
Seasonal_Index	dim_seasonal_index	Dimension	→ Commodity, Month (1–12)
MSP_vs_Market	fact_msp_analysis	Fact	→ Commodity, Season
Statewise_Prices	fact_state_prices	Fact	→ State, Commodity, Period
Price_Forecast_30Day	fact_forecasts	Fact	→ Commodity, Date
KPI_Summary	dim_kpis	KPI Source	Used in Card visuals
Arrivals_Breakdown	fact_arrivals	Fact	→ Commodity, Group

9.2 Recommended DAX Measures

<pre>-- Avg Modal Price (filtered) Avg_Modal_Price = AVERAGE(fact_mandi_prices[Modal_Price_Rs]) -- MSP gap (negative = below MSP) MSP_Gap_Pct = DIVIDE([Avg_Modal_Price] - MAX(fact_msp_analysis[MSP_Rs_Qtl]), MAX(fact_msp_analysis[MSP_Rs_Qtl]), 0) * 100 -- Total arrivals (all markets) Total_Arrivals = SUM(fact_arrivals[Total_Arrivals]) -- Forecast vs current price change % Forecast_Change_Pct = DIVIDE([Forecast_7d] - [Current_Price], [Current_Price], 0) * 100</pre>

9.3 Dashboard Pages

Page	Visuals	Primary Data	Key Filters
Overview	KPI Cards · Donut chart · Bar chart	KPI_Summary + Arrivals_Breakdown	Date, Commodity Group

Mandi Prices	Matrix table · Scatter plot · Heatmap	Mandi_Prices	Date, Group, Market, MSP Status
Seasonal Index	Heatmap (12×16) · Line chart	Seasonal_Index	Commodity, Month range
30-Day Forecast	Line+band chart · Column chart · Table	Price_Forecast_30Day	Commodity, Model
Top States	Map visual · Bar compare · Column chart	Statewise_Prices	Commodity, Period
MSP Analysis	Waterfall chart · Risk table · Gauge	MSP_vs_Market	Season, Commodity, Status

10. Conclusion & Future Work

10.1 What We Achieved

Data Sources	Crops Analysed	SQL Queries	Forecast Accuracy	Dashboard Pages
4 <i>Agmarknet+eNAM+Govt+FAO</i>	33 <i>16 with seasonal index</i>	10 <i>+ 3 mat. views</i>	91.4% <i>Ensemble MAPE 8.6%</i>	6 <i>Power BI ready</i>

10.2 Recommendations

Priority	Action	Owner	Timeline
URGENT	Initiate Groundnut MSP procurement — ₹49.5 Cr income risk	NAFED / State Agri Dept	Oct 2025
HIGH	Deploy lean-season advisory for Tomato farmers (Jun–Aug premium +25%)	FPO / Kisan Portal	Jun 2026
HIGH	Integrate IMD rainfall data as exogenous variable in Onion/Tomato models	Data Team	Q3 2026
MEDIUM	Expand to 500+ APMCs (from 42) for national-level intelligence	Agmarknet API Team	Q4 2026
MEDIUM	Automate daily data ingestion pipeline with Airflow DAG	Data Engineering	Q3 2026
LOW	Add NDVI satellite data for crop health → yield → price correlation	Research Team	2027

10.3 Project Impact

Quantified Impact • ₹49.5 Cr farmer income at risk identified (Groundnut) • 30-day advance forecast → better sell-timing for farmers • Seasonal index enables 15–56% lean-season premium capture	Future Revenue Potential • B2G: State agri dept licensing (₹50L–₹2Cr/yr) • B2B: Commodity trader subscriptions (₹10–50K/mo) • B2C: Kisan app integration (freemium model) • API licensing to agritech platforms
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• 95 commodities × 42 markets monitored daily

Agri Commodity Price Forecasting & Market Intelligence — End of Case Study